

Fill a Seven Day Pill Organizer SOP

One episode fills a seven day pill organizer from four bottles, following a printed dose chart. The organizer is standing at the center of the desk with all seven lids shut when recording starts, the four bottles stand capped in the bottle row in one corner of the desk, and the chart lies flat at the chart spot along the back edge. All seven lids are opened first. Then each bottle in turn is taken to the dispense spot, its cap is taken off, its tablets are put in the compartments the chart calls for, its cap goes back on, and it returns to its slot. When all four bottles are back, the seven lids are closed and the filled row is checked against the chart through the clear lids.

The order never changes. Seven lids open, then bottle A, then B, then C, then D, then seven lids closed, then the check. Within one bottle the compartments are filled in day order, Monday first through Sunday last. A bottle is finished and capped before the next one is touched.

Four rules never bend. Only one bottle is open at any moment. An open bottle stands at the dispense spot and is never lifted, carried, tilted, or moved while its cap is off. Tablets leave a bottle one at a time, taken by a gripper, and are never poured, tipped, or shaken out. A tablet that has left its bottle never goes back into a bottle, whatever happens to it.

The right gripper does the primary work: it takes every cap off and puts every cap on, fills Thursday through Sunday, opens and closes the Thursday through Sunday lids, and carries every bottle out and back when the bottle row is on the right. The left gripper supports: it fills Monday through Wednesday, opens and closes the Monday through Wednesday lids, holds the bottle while the right gripper works its cap, pins the organizer while the right gripper works a lid, and carries every bottle out and back when the bottle row is on the left. Neither gripper takes the other's jobs.

The desk is set up in one of four ways. Only the bottle row moves; the organizer, the dispense spot, the cap spot, the lid band, and the discard cup are in the same place in all four. Both center zones are taken, by the dispense spot at the back and the discard cup at the front, so the bottle row uses the four corners.

- **Config L1:** the bottle row is at the back left, and the chart lies at the back right.
- **Config L2:** the bottle row is at the front left.
- **Config R1:** the bottle row is at the back right.
- **Config R2:** the bottle row is at the front right.

Where a step depends on the setup it says so on an **IF** line — look at the desk and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the bottle row starts at the back left (**Config L1**), the front left (**Config L2**), the back right (**Config R1**), or the front right (**Config R2**). One config per episode, chosen

before recording and never changed mid-episode.

- **Same-side rule:** the gripper on the bottle row's side carries each bottle out to the dispense spot and back to its slot — the left gripper in Config L1 and L2, the right gripper in Config R1 and R2. No arm reaches across the desk for a bottle. Both grippers reach the dispense spot, so nothing is ever handed over.
- **Fixed roles:** everything else is the same in all four configs — the right gripper works every cap at the cap spot, fills Thursday through Sunday, and works their lids; the left gripper fills Monday through Wednesday, works their lids, holds the bottle while its cap is worked, and pins the organizer while the right gripper works a lid.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole desk: the chart spot at the back left (the back right in Config L1), the dispense spot at the back center, the bottle row in this episode's corner, the cap spot on the right side, the organizer at the center, the lid band in front of it, and the discard cup at the front center.
3. The overhead view reads the printed chart cell by cell, reads the day name printed on every compartment, and reads the inside of every compartment, so the count and the color of the tablets in each one can be told apart with the lids open and again with the lids closed.
4. Both arms are at home with grippers open.
5. The desk is clear of anything but the chart, the organizer, the four bottles, and the discard cup.
6. The right gripper reaches (Config R1 and R2) all four bottle slots, the dispense spot, the cap spot, the Thursday, Friday, Saturday, and Sunday compartments and their lids, the organizer back rail at its right end, and the discard cup, without the arm leaning out or reaching a joint limit.
7. The left gripper reaches (Config L1 and L2) all four bottle slots, the dispense spot, the Monday, Tuesday, and Wednesday compartments and their lids, the organizer back rail at its left end, and the discard cup, without the arm leaning out or reaching a joint limit.

Materials checklist

1. One **pill organizer** stands at the organizer spot at the center of the desk, square to the desk, long side parallel to the front edge. It has seven compartments in one row, printed with their day names and running **Monday at the left end through Sunday at the right end**. Every compartment is empty and every lid is shut and seated.
2. Each compartment has its own clear **lid**, hinged along the organizer's front long edge, so the lid swings toward the front edge to open and lies flat on the desk in front of its compartment. Every hinge is sound and every lid stays where it is put.
3. Each lid seats under a steady press and stays down on its own. No lid needs a turn, a catch, or a second hand to hold it shut.
4. The organizer sits firmly enough that a gripper pressing a lid shut does not slide it.
5. Four **bottles** stand capped in the **bottle row** in the corner for this episode's config, in a line running from the desk midline toward the near side edge: **bottle A** nearest the midline, then **B**, then **C**, then **D** nearest the side edge. Each bottle is printed with its letter in large type on the side facing the front edge.
 - **Config L1**: back left, where the chart spot otherwise is
 - **Config L2**: front left, clear of the lid band and the discard cup
 - **Config R1**: back right, right of the dispense spot
 - **Config R2**: front right, clear of the cap spot, the lid band, and the discard cup
6. Each bottle is short, flat based, and stands on its own without leaning. Its body is narrow enough for the gripper to close on it and lift it without the bottle turning in the fingers.
7. The mouth of each bottle is wide enough that a gripper comes straight down inside it and closes on one tablet without touching the rim.
8. Each **cap** lifts straight off its bottle and presses straight back on. No cap turns, screws, or needs force to lift.
9. Each bottle holds **tablets** of one color only, in a single loose layer resting on the bottle floor, never stacked or wedged: bottle A red, bottle B blue, bottle C white, bottle D yellow. The four colors are clearly different from each other and from the organizer.
10. The tablets are plain inert tablets used for data collection. No real medicine is on the desk.
11. Every tablet is whole, dry, and loose, large enough for a gripper to take one on its own, and small enough that two of them sit in a compartment without touching a lid.
12. Each bottle holds at least four more tablets than the chart calls for, so a dropped tablet never empties a bottle mid episode.
13. One printed **dose chart** lies flat at the **chart spot** at the back left (the back right in Config L1), square to the desk, printed side up. It has four rows marked A, B, C, D, one per bottle, and seven columns marked **Monday through Sunday** in the same order as the compartments, one per compartment. Every cell gives a count of **0, 1, or 2**, and the counts across the whole chart add up to no more than 20 tablets.

14. The chart is fixed to the desk or weighted so a passing arm cannot move it. Neither gripper ever touches it.
15. One **discard cup** stands empty at the front center, forward of the lid band, heavy enough that a gripper lowering a tablet into it does not tip it.
16. The **dispense spot** at the back center, the **cap spot** on the right side, and the **lid band** in front of the organizer are all bare.

Workspace layout

- **Chart spot:** back left, or back right in Config L1 when the bottle row takes the back left. The printed dose chart lies flat here all episode and is never touched by either gripper.
- **Dispense spot:** back center, on the desk midline, just behind the organizer. One open bottle at a time stands here. Both grippers reach into it.
- **Bottle row:** back left (**Config L1**), front left (**Config L2**), back right (**Config R1**), or front right (**Config R2**). The four capped bottles stand here in order A nearest the desk midline through D nearest the side edge. The left gripper's zone in Config L1 and L2, the right gripper's in Config R1 and R2.
- **Cap spot:** the right side of the desk, right of the organizer's right end and clear of the bottle row and the lid band. One cap at a time lies here, open side up. The right gripper's zone.
- **Organizer spot:** center of the desk, in front of the dispense spot. Monday is at the left end and Sunday at the right end. Monday, Tuesday, and Wednesday are the left gripper's compartments. Thursday, Friday, Saturday, and Sunday are the right gripper's.
- **Lid band:** the strip of bare desk between the organizer and the front edge, where the seven open lids lie. Nothing else is ever put there.
- **Discard cup:** front center, forward of the lid band. Takes every tablet that leaves a bottle and does not land in the compartment it was meant for. Both grippers reach it.

Everything the right gripper carries stays on the right or at the center. The left gripper carries nothing further right than the Wednesday compartment and the discard cup. In Config L1 and L2 the left gripper also carries every bottle between the bottle row and the dispense spot, and the right gripper carries only caps. Nothing is ever carried over the open bottle.

Vocabulary

- **Arm assignment:** the right gripper works every cap, fills Thursday through Sunday, works the Thursday through Sunday lids, and carries every bottle in Config R1 and R2. The left gripper fills Monday through Wednesday, works the Monday through Wednesday lids, holds the bottle while a cap comes off or goes on, pins the organizer while the right gripper works a lid, and carries every bottle in Config L1 and L2. Neither gripper takes the other's jobs.
- **Compartment:** one of the seven wells in the organizer, printed with its day name, running Monday at the left end through Sunday at the right end.

- **Day order:** Monday first, then Tuesday, and so on through Sunday. Every fill runs in day order, left to right along the organizer.
- **Bottle order:** bottle A first, then B, then C, then D. Every episode runs in bottle order.
- **Bottle row:** where the four capped bottles stand at the start and the end — back left (**Config L1**), front left (**Config L2**), back right (**Config R1**), or front right (**Config R2**). One per episode, chosen before recording and never changed mid-episode.
- **Dose cell:** the one cell on the chart where the current bottle's row meets the current day's column. It gives 0, 1, or 2 and is the only thing that sets how many tablets go in.
- **Open bottle:** a bottle with its cap off. It stands at the dispense spot and nowhere else, and it is never lifted, carried, tilted, or slid while it is open.
- **Capped bottle:** a bottle with its cap pressed on and seated flat all round. Only a capped bottle is ever carried.
- **Seated cap:** the cap sits flat all round on the bottle mouth with no gap and no tilt, and stays on when the gripper lifts clear.
- **Seated lid:** the lid lies shut and flat on its compartment, level with the lids beside it, and stays down when the gripper lifts clear.
- **Open lid:** the lid swung right over its hinge and lying flat on the desk in the lid band, clear of its own compartment and of the compartments beside it.
- **Stray tablet:** any tablet that has left a bottle and is not resting in the compartment it was meant for. It goes in the discard cup. It never goes into any compartment and never goes back into any bottle.
- **Organizer back rail:** the strip of the organizer body behind the row of compartments. It is the only part of the organizer a gripper pins.
- **Lift:** coming straight down onto the grasp point, closing, and raising straight up until there is daylight under the thing, before anything moves sideways.
- **Hold:** a gripper closed on something standing on the desk, keeping it still, without lifting it and without dragging it.
- **Pin:** a closed gripper pressed straight down to hold something still. A pin never pinches and never travels.
- **Press:** the closed gripper set straight down, held still for 2 seconds, and lifted straight off. A press never slides and never rubs.
- **Fold:** a gripper pinching a lid at the middle of its free edge, swinging it over its hinge until it lies flat, holding for 2 seconds, and releasing.
- **Settled:** the thing stays put after the gripper lifts clear, with nothing springing back, rolling, or tipping.
- **Release point:** a compartment, the dispense spot, a bottle slot, the cap spot, the lid band, or the discard cup. A gripper opens nowhere else.

Steps

Step 1 opens all seven lids and runs once. Steps 2 to 4 are one bottle. Run them in full four times, in bottle order A, B, C, D. Steps 5 to 7 run once, after all four bottles are back in the bottle row. Only the first pick in Step 2 and the return in Step 4 depend on the config: the gripper on the bottle row's side carries the bottle out to the dispense spot and back (left in Config L1 and L2, right in Config R1 and R2). Every other line is the same in all four configs.

Step 1: Open all seven lids

Goal: all seven lids lying open and flat in the lid band, with all seven compartments empty and clear from above.

- The **right gripper**, closed, **pins** the **organizer back rail** at its right end and holds it there while the left gripper works.
- The **left gripper** **folds** the **Monday** lid open: it pinches the lid at the middle of its free edge, swings it toward the front edge and over its hinge until it lies flat on the desk in the lid band, holds it for 2 seconds, and releases.
- The **left gripper** opens the **Tuesday** lid, then the **Wednesday** lid, the same way.
- The **left gripper**, closed, **pins** the organizer back rail at its left end. Only once that pin is down does the right gripper lift clear.
- The **right gripper** opens the **Thursday**, **Friday**, **Saturday**, and **Sunday** lids in that order, the same way.
- Both grippers lift clear.

The organizer is pinned at the far end from the lid being worked. A lid swung open on an unpinned organizer drags the whole row toward the front edge and puts every compartment out of place.

Check: all seven lids lie flat and open in the lid band, none standing part way up, none folded back over its own compartment, and none lying over a compartment beside it. All seven compartments are empty and open from above. The organizer still stands square at the organizer spot with its row running left to right and Monday at the left end. If a lid sprang part way back, fold it open again and hold it longer before releasing. If the organizer shifted, stop, log it as a system issue, and discard the episode.

Step 2: Take the next bottle to the dispense spot and take its cap off

Goal: the next bottle in order standing open at the dispense spot, with its cap lying open side up at the cap spot.

Look where the bottle row is before reaching for the bottle.

- **IF the bottle row is on the left (Config L1 or L2):** the **left gripper** closes on the next bottle at the middle of its body in the **bottle row**, **lifts** it straight up until its base is clear of the

desk, carries it upright to the **dispense spot**, lowers it until its base rests flat, and releases once it is settled. The bottle is never tilted on the way.

- **IF the bottle row is on the right (Config R1 or R2):** the **right gripper** takes the next bottle to the dispense spot the same way.

Then, in all four:

- The **left gripper** closes on the bottle at the middle of its body and **holds** it while the cap comes off.
- The **right gripper** closes on the **cap** at its side, **lifts** it straight up off the mouth, carries it level to the **cap spot**, lowers it **open side up**, and releases.
- Both grippers lift clear.

The bottle is held while the cap comes off. A cap pulled off a loose bottle lifts the bottle with it and drops it back onto the desk on its rim.

Check: the bottle stands upright and settled at the dispense spot with its letter facing the front edge, its mouth open, and its tablets lying loose inside. The cap lies at the cap spot, open side up, touching nothing else. No other bottle is open and the other three still stand capped in their own slots in the bottle row. No tablet has left the bottle. If the bottle landed leaning or off the dispense spot, hold it, lift it clear, and set it down again before the cap comes off.

Step 3: Fill the compartments this bottle's chart row calls for

Goal: every compartment holds the number of this bottle's tablets that its dose cell gives, and nothing else has changed.

- The **right gripper** closes on the open bottle at the middle of its body and **holds** it at the dispense spot while the left gripper works.
- Work **Monday, Tuesday, and Wednesday** in day order with the **left gripper**. For each one, read the **dose cell** where this bottle's row meets that day's column. If it gives 0, leave that compartment alone and move to the next. If it gives 1 or 2, take that many tablets, one at a time.
- For each tablet, the **left gripper** comes straight down inside the open bottle, closes on **one** tablet, and **lifts** it straight up until it is clear of the rim. It carries the tablet level to the open compartment, lowers it until it rests on the compartment floor, releases, and lifts clear.
- When Monday, Tuesday, and Wednesday are done, the **left gripper** closes on the bottle at the middle of its body and holds it. Only once that hold is on does the right gripper lift clear.
- Work **Thursday, Friday, Saturday, and Sunday** in day order with the **right gripper**, one tablet at a time, the same way.
- Both grippers lift clear.

Tablets are picked one at a time out of a bottle that another gripper is holding. A bottle tipped over a compartment to shake tablets out puts an unknown number in, spills the rest across the

desk, and cannot be counted back.

Stray branch: if a tablet falls, lands on a lid, lands on the desk, or lands in the wrong compartment, the gripper that dropped it stops, picks that tablet up, carries it to the **discard cup**, lowers it in, and releases, before it takes anything else. It never goes into a compartment and never goes back into any bottle. Then take a fresh tablet from the bottle and carry on. If the tablet has rolled somewhere that gripper cannot reach without leaning out, leave it, log a dropped tablet, and carry on.

Check: each compartment from Monday to Sunday holds exactly the number of this bottle's tablets that its dose cell gives, and a compartment whose cell gives 0 holds none of this color. Tablets put in by earlier bottles are still there, still in their own compartments, and none have been lifted, pushed, or moved. No tablet rests on a lid, on the organizer body, or loose on the desk. The bottle still stands upright and settled at the dispense spot with tablets left inside, and the cap still lies at the cap spot. If a compartment holds too many of this color, lift the extras out to the discard cup and take fresh count from the chart. If it holds too few, take the missing tablets now.

Step 4: Put the cap back on and return the bottle

Goal: the bottle standing capped in its own slot in the bottle row, with the dispense spot and the cap spot bare.

- The **left gripper** closes on the bottle at the middle of its body at the dispense spot and **holds** it.
- The **right gripper** closes on the **cap** at its side at the cap spot, **lifts** it straight up, and carries it level until it is centered over the bottle mouth.
- The **right gripper** lowers the cap straight down onto the mouth and **presses** it down until it sits flat all round, holds for 2 seconds, and releases.
- The **right gripper** lifts clear, then the left gripper lifts clear.
- **IF Config L1 or L2:** the **left gripper** closes on the capped bottle at the middle of its body, **lifts** it straight up, carries it upright to its own slot in the **bottle row**, lowers it until its base rests flat, and releases once it is settled. **IF Config R1 or R2:** the **right gripper** returns it the same way.
- That gripper lifts clear.

The cap goes on before the bottle moves. An open bottle carried across the desk tips its tablets out over the organizer and over the open lids.

Check: the cap sits flat all round on the bottle with no gap and no tilt, and stays on when the gripper lifts clear. The bottle stands upright and settled in its own slot with its letter facing the front edge and the row still in order A, B, C, D. The dispense spot and the cap spot are bare and no tablet lies loose. If the cap sits tilted or lifts with the gripper, hold the bottle, lower the cap onto the mouth again, and press it again before the bottle is carried.

Run Steps 2 to 4 again for bottle B, then bottle C, then bottle D. Go on to Step 5 only when all four bottles stand capped in the bottle row and the dispense spot and the cap spot are bare.

Step 5: Close all seven lids

Goal: all seven lids shut and seated, with the lid band bare.

- The **right gripper**, closed, **pins** the **organizer back rail** at its right end and holds it there while the left gripper works.
- The **left gripper** pinches the open **Monday** lid at the middle of its free edge, swings it away from the front edge and over its hinge until it lies on the compartment, and **presses** it down until it seats flat, holding for 2 seconds before releasing.
- The **left gripper** closes the **Tuesday** lid, then the **Wednesday** lid, the same way.
- The **left gripper**, closed, **pins** the organizer back rail at its left end. Only once that pin is down does the right gripper lift clear.
- The **right gripper** closes the **Thursday**, **Friday**, **Saturday**, and **Sunday** lids in that order, the same way.
- Both grippers lift clear.

Each lid is swung over and pressed down. A lid pushed shut from the side rides up on the rim, drags tablets to one end of the compartment, and does not seat.

Check: all seven lids are shut, seated flat, and level with each other, none standing part way up and none sprung back open. No tablet is caught under a lid edge or trapped between the lid and the rim. The lid band is bare and the organizer still stands square at the organizer spot. If a lid sprang open, press it again and hold it longer before releasing.

Step 6: Check the filled row against the chart

Goal: every one of the seven compartments matches its column on the chart, read through the clear lids.

- Go along the organizer from **Monday to Sunday** in day order. For each compartment, count the tablets of each color through its clear lid and check them against that day's column on the chart, row A through row D.
- Confirm no tablet lies on a lid, on the organizer body, or loose on the desk.
- Confirm all four bottles stand capped in the bottle row in order A, B, C, D, the chart still lies flat and square at the chart spot, and the dispense spot, the cap spot, and the lid band are bare.

Correction branch: if a compartment does not match its column, the gripper that owns that day opens its lid again as in Step 1, with the other gripper pinning the back rail. For a tablet that should not be there, that gripper lifts it out, carries it to the **discard cup**, lowers it in, and

releases. For a missing tablet, run Steps 2 to 4 again for that bottle and put only the missing tablets in that compartment. Then close the lid again as in Step 5 and check that day again.

Check: all seven compartments match the chart, all seven lids are shut and seated, and nothing loose is on the desk.

Step 7: End the episode

- Confirm Step 6 passed on all seven compartments.
- Return both arms home, clear of the organizer, the bottle row, and the discard cup, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Open all seven lids and tip each compartment out. Sort the tablets by color and return each color to its own bottle. Never put a tablet in a bottle of another color.
2. Empty the discard cup and throw those tablets away. Tablets that reached the discard cup are never reused. Confirm the cup is empty and standing at the front center.
3. Pick up any tablet left loose on the desk and throw it away.
4. Check every tablet going back into a bottle. Throw away any that is chipped, split, damp, or stuck to another, and top each bottle back up so it holds at least four more than the next chart calls for.
5. Wipe each compartment dry, close all seven lids, and confirm each one seats flat. Replace the organizer if a hinge is cracked or a lid will not stay open or will not seat.
6. Press every cap on and stand the four bottles in the bottle row for the next episode's config — back left (Config L1), front left (Config L2), back right (Config R1), or front right (Config R2) — in order A, B, C, D from the desk midline outward, letters facing the front edge. Replace a cap that will not lift off cleanly or will not seat flat.
7. Lay the next printed dose chart flat at the chart spot, the back left or (Config L1) the back right, printed side up, and fix or weight it. Confirm its columns run Monday through Sunday in the same order as the compartments, that every cell gives 0, 1, or 2, and that the whole chart adds up to no more than 20 tablets.
8. Confirm the organizer stands square with Monday at the left end, and that the dispense spot, the cap spot, and the lid band are bare. Wipe the desk.
9. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the start position: the violations below were written for Config R1 (bottle row at the back right, chart at the back left). The pickup and arm-role cues will be rewritten later to cover all four start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the desk — the bottle row is not in the corner for the config; a gripper reaches across the desk for a bottle or carries a bottle from the far side; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the gripper on the bottle row's side carries each bottle out to the dispense spot and back, left in Config L1 and L2 and right in Config R1 and R2; no arm reaches across the desk; the IF line followed is the one for the config on the desk).
- **Coaching note:** look where the bottle row is before the first reach, then follow that config's IF lines through Steps 2 and 4.

Violation: Filling started before all seven lids were open

- **Visible cue:** a bottle is lifted from the bottle row, or a tablet is put in a compartment, while any lid is still shut or standing part way up.
- **SOP rule broken:** Step 1, all seven lids lie open and flat in the lid band before Step 2 begins.
- **Coaching note:** all seven lids open first. Then the first bottle.

Violation: Bottles worked out of order

- **Visible cue:** bottle B, C, or D is carried to the dispense spot before bottle A has been filled, capped, and returned, or the order A, B, C, D is broken anywhere.
- **SOP rule broken:** Steps 2 to 4, run one whole bottle at a time in the order A, B, C, D.
- **Coaching note:** A, then B, then C, then D. Never jump ahead.

Violation: More than one bottle open

- **Visible cue:** a second cap is lifted while another bottle stands open, or two bottles stand uncapped at once anywhere on the desk.
- **SOP rule broken:** Steps 2 and 4, a bottle is capped and back in its slot before the next cap is touched.
- **Coaching note:** one open bottle on the desk, ever.

Violation: Open bottle lifted or moved

- **Visible cue:** a bottle with its cap off is lifted, carried, tilted, slid, or nudged off the dispense spot.
- **SOP rule broken:** Steps 2 to 4, an open bottle stands still at the dispense spot and only a capped bottle is carried.
- **Coaching note:** cap on before it moves.

Violation: Tablets poured, tipped, or shaken out

- **Visible cue:** the bottle is tilted, shaken, tapped, or upended over a compartment, the desk, or a hand, instead of a gripper reaching in and taking one tablet.
- **SOP rule broken:** Step 3, each tablet is taken out of the bottle by a gripper, one at a time.
- **Coaching note:** reach in and take one. Never tip the bottle.

Violation: More than one tablet in a pick

- **Visible cue:** a gripper lifts two or more tablets out of the bottle together, or carries two to a compartment in one trip.
- **SOP rule broken:** Step 3, one tablet per pick and per carry.
- **Coaching note:** one at a time, even when the cell says two.

Violation: Bottle not held while worked

- **Visible cue:** a cap is lifted off or pressed on with no gripper holding the bottle, or a gripper reaches into the open bottle while the other gripper is off it, and the bottle rocks, turns, slides, or tips.
- **SOP rule broken:** Steps 2, 3, and 4, one gripper holds the bottle at the middle of its body whenever the other gripper works its cap or reaches inside it.
- **Coaching note:** hold it first. Then work it.

Violation: Wrong count for a day

- **Visible cue:** a compartment ends the episode with more or fewer tablets of a color than its dose cell gives, or tablets are put in a compartment whose cell gives 0.
- **SOP rule broken:** Step 3, the dose cell where the bottle's row meets that day's column is the only thing that sets how many tablets go in.
- **Coaching note:** read the cell before every day. Zero means skip it.

Violation: Wrong day or wrong order

- **Visible cue:** a tablet goes into a compartment other than the day being worked, or the days are filled out of order rather than Monday through Sunday.
- **SOP rule broken:** Step 3, fill in day order, Monday first through Sunday last.
- **Coaching note:** left to right, Monday to Sunday, one day at a time.

Violation: Tablet put back in a bottle

- **Visible cue:** a tablet that has left a bottle is dropped, lowered, or brushed back into any bottle, including the one it came from.
- **SOP rule broken:** Step 3 stray branch, a tablet that has left a bottle goes in the discard cup and never back into a bottle.
- **Coaching note:** once it is out, it is out. Discard cup or nowhere.

Violation: Stray tablet left or reused

- **Visible cue:** a tablet that fell, landed on a lid, or landed in the wrong compartment is left where it lies while the work carries on, is pushed into a compartment, or is picked up and used as the dose tablet.
- **SOP rule broken:** Step 3 stray branch, the gripper that dropped it clears it to the discard cup before it takes anything else, and a fresh tablet is taken from the bottle.
- **Coaching note:** stop, clear it to the cup, then take a fresh one.

Violation: Tablets already placed disturbed

- **Visible cue:** a gripper lifts, pushes, drags, or knocks a tablet put in by an earlier bottle, or a tablet moves from one compartment into another.
- **SOP rule broken:** Steps 3 and 5, tablets already in a compartment stay where they are.
- **Coaching note:** come down into an empty part of the compartment and leave the rest alone.

Violation: Organizer not pinned while a lid is worked

- **Visible cue:** a lid is opened or pressed shut with no gripper on the organizer back rail, or the pinning gripper lifts away before the other is clear, and the organizer slides, turns, or shifts off its spot.
- **SOP rule broken:** Steps 1 and 5, one gripper pins the back rail at the far end while the other works a lid, and lifts clear only after that gripper is off.
- **Coaching note:** pin first, let go last.

Violation: Lid worked wrongly

- **Visible cue:** a lid is pushed shut from the side, forced, bent, twisted, prised, swung away from the front edge to open, left standing part way up, folded back over its own

compartment, or left lying over a compartment beside it.

- **SOP rule broken:** Steps 1 and 5, a lid is pinched at the middle of its free edge, swung over its hinge, and laid flat or pressed until it seats.
- **Coaching note:** pinch the free edge, swing it over, hold it down.

Violation: Cap set down or seated wrongly

- **Visible cue:** the cap is laid open side down, dropped on the desk, left at the dispense spot or anywhere but the cap spot, put on a bottle other than its own, or released tilted, sitting proud, or lifting away with the gripper.
- **SOP rule broken:** Steps 2 and 4, the cap lies open side up at the cap spot and goes back on its own bottle pressed flat all round.
- **Coaching note:** open side up on the spot, flat all round going back on.

Violation: Lids closed before all four bottles were done

- **Visible cue:** a lid is pressed shut while a bottle is still open, still at the dispense spot, or still to be run.
- **SOP rule broken:** Step 5, all four bottles are capped and back in the bottle row before any lid is closed.
- **Coaching note:** four bottles home first, then close the lids.

Violation: Chart touched or moved

- **Visible cue:** a gripper contacts, lifts, slides, or covers the printed chart, or the chart ends the episode away from the chart spot or not square to the desk.
- **SOP rule broken:** Steps 3 and 6, the chart is read where it lies and never touched.
- **Coaching note:** read it. Never reach for it.

Violation: Arms swapped roles

- **Visible cue:** the left gripper carries a bottle, works a cap, or works the Thursday, Friday, Saturday, or Sunday compartment or its lid; or the right gripper works the Monday, Tuesday, or Wednesday compartment or its lid other than in the Step 6 correction branch.
- **SOP rule broken:** Steps 1 to 6, the right gripper takes the bottles, the caps, and Thursday through Sunday, and the left gripper takes Monday through Wednesday and the holds and pins.
- **Coaching note:** right takes the bottles and Thursday to Sunday. Left takes Monday to Wednesday and holds.

Violation: Regripped in the air

- **Visible cue:** a gripper shifts, rolls, or seats its hold again on a bottle, a cap, a tablet, or a lid without setting it down first.

- **SOP rule broken:** Steps 1 to 6, take a fresh hold only after the thing is resting.
- **Coaching note:** put it down before you take a new hold.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens above a compartment, a bottle, or the discard cup and the thing falls the rest of the way, or it opens over bare desk, over open space, or over the wrong zone.
- **SOP rule broken:** Steps 1 to 6, the gripper opens at a compartment, the dispense spot, a bottle slot, the cap spot, the lid band, or the discard cup, once the thing is resting, and nowhere else.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Placement corrected by shoving

- **Visible cue:** a gripper nudges, pats, brushes, or straightens a bottle, a cap, a tablet, a lid, or the organizer after releasing it, instead of taking hold of it again and setting it down properly.
- **SOP rule broken:** Steps 1 to 6, correct a bad placement by taking hold of the thing and setting it down again.
- **Coaching note:** pick it up and set it again, or leave it alone.

Violation: Item dropped, collided, or knocked off the desk

- **Visible cue:** a bottle, a cap, a tablet, or the organizer falls off the desk or is knocked over; the discard cup or a bottle in the row is struck and shifts; or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 6, carry on a clear path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Check skipped or a mismatch left standing

- **Visible cue:** the arms go home without the seven compartments being read against the chart, or a day that does not match its column is left as it is.
- **SOP rule broken:** Steps 6 and 7, read Monday through Sunday against the chart and run the correction branch on any day that does not match before the arms go home.
- **Coaching note:** read all seven days against the chart. Fix what does not match.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a lid open or not seated, a bottle uncapped or out of its slot, a cap at the cap spot, a tablet loose on the desk or on a lid, or an arm away from home.
- **SOP rule broken:** Step 7, confirm the check passed, then return both arms home, then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective cap:** it will not lift off under a straight pull, or it will not seat flat under a correct press.
- **Defective lid or hinge:** a lid will not stay open, will not seat shut, is cracked, or its hinge splits during the episode.
- **Bottle holds fewer tablets than its chart row calls for**, so the row cannot be filled.
- **Defective tablets:** found broken, chipped, damp, or stuck together in a bottle that has not been worked, or so large that two will not sit in a compartment under a closed lid.
- **Tablets wedged or stacked in the bottle**, so a gripper cannot take one on its own.
- **Chart unreadable or wrong:** a cell is blank, smudged, or unreadable overhead, a column is missing or out of day order, or a cell gives a count outside 0 to 2.
- **Wrong tablet color found in a bottle** that has not been worked.
- **Day names on the organizer unreadable overhead**, or the organizer is printed in an order other than Monday at the left end through Sunday at the right end.
- **The organizer, a bottle, the chart, or the discard cup shifts out of place** on its own during the episode.

Annotation subtasks (from SOP)

1. Open all seven lids
2. Take the next bottle to the dispense spot and take its cap off
3. Fill the compartments this bottle's chart row calls for
4. Put the cap back on and return the bottle to its slot
5. Close all seven lids
6. Check the filled row against the chart
7. Return both arms home and end the episode